Do Now · retrieval

Lesson 149 · Client–Server and Peer-to-Peer Network Models

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which device connects several wireless users to an existing school LAN?

Q2

Which storage type is volatile?

Q3

Which layer in TCP/IP handles IP routing?

Q4

What data structure follows FIFO?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Client–server — Many clients request services from a managed central server.

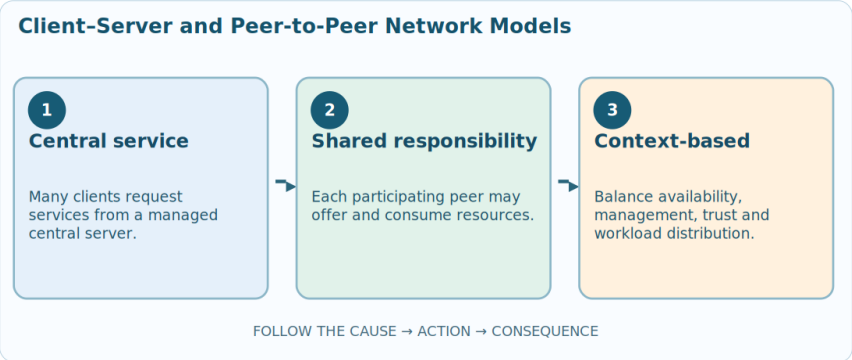
Core idea 2

Peer-to-peer — Each participating peer may offer and consume resources.

Core idea 3

Select architecture — Balance availability, management, trust and workload distribution.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

client–server — Managed service provided by server to requesting clients.

peer-to-peer — Peers directly share services/resources.

centralisation — Services controlled from a smaller set of authorities.

scalability — How well a system handles growth in load/users.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A school stores assessment marks requiring access control, backups and a shared authoritative copy. Recommend a model.

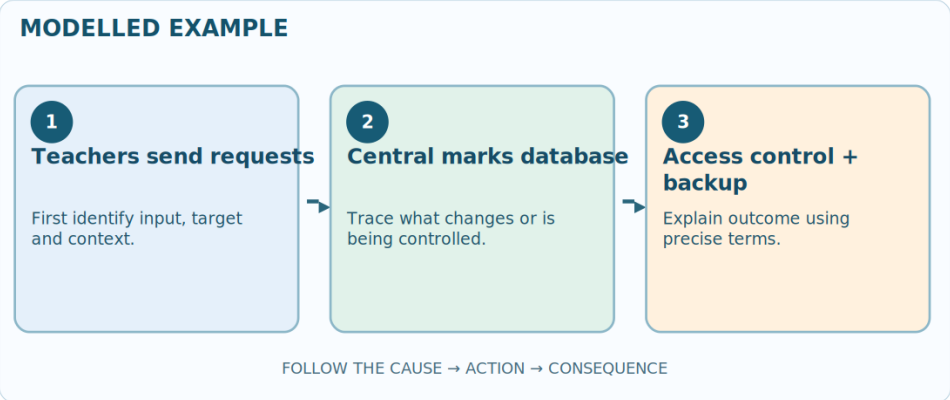
Follow the worked method

1. Teachers send requests — Identify the facts and make an initial prediction.

2. Central marks database — Explain the mechanism or trace the intermediate stage.

3. Access control + backup — Client–server suits managed central data, controlled permissions, auditing and backups. A server failure can affect many clients, so redundancy and resilient backups matter. Peer-to-peer file sharing is harder to manage consistently.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Peers hold file blocks
Identify the facts and make an initial prediction.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define client, server and peer in the context of a network service. [6]

Question 2

Explain two benefits and one limitation of client–server for a school's student database. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare P2P and client–server for distribution of a large video archive, including integrity checks. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A network using client–server can never contain any peer-to-peer activity.

Use key vocabulary from memory

1. Explain client–server in your own words.

2. Explain peer-to-peer in your own words.

3. Explain centralisation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.